

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 2.1: Properties of Waves — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Properties of Waves
Driving Question	DRIVING QUESTION: "Why does the matatu's horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel — and what do these two mysteries reveal about how ALL waves behave?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon Launch — "The Matatu Horn and the Fading Radio"	Students observed (via live buzzer demonstration and PhET Sound simulation) that a moving sound source produces a pitch change — higher as it approaches, lower as it moves away. They also observed from the tunnel diagram that FM waves (~2.9 m wavelength) are smaller than the Kabete tunnel width (~8 m), while AM waves (~529 m) are far larger than the tunnel.	Students learned that sound and radio signals are both wave phenomena, and that waves transfer energy without transferring matter. They learned that a wave can be represented as a drawing showing direction of travel, and that properties such as wavelength may be relevant to how a wave interacts with a physical barrier.	Students are not yet able to fully explain either part of the phenomenon — this is intentional. Partial emerging explanations include: (a) the pitch change is related to the motion of the matatu relative to the listener, and (b) the FM blackout may be related to the size of the wave relative to the size of the tunnel opening. Both explanations remain incomplete and will be refined across the unit.
Lesson 2 What Is a Wave? Building Our First Model from Everyday Kenyan Phenomena	Students listened to an audio clip of the Doppler-shifted matatu horn and the AM/FM Kabete tunnel narration. They observed ocean wave images (Nyali Beach, Mombasa), a stadium crowd wave (Kasarani Stadium), and chai cup ripples. They used the PhET Sound Simulation to watch individual air particles move back and forth while the wave disturbance travels forward.	A wave is a disturbance that transfers energy from one point to another without transferring matter. Transverse waves have crests, troughs, amplitude, wavelength, a rest position, and a direction of travel distinct from the direction of particle vibration. Both the matatu horn sound and the Classic 105 FM / AM radio signals are waves — disturbances that carry energy from a source to a receiver.	Students began to explain why the horn's pitch change and the FM/AM tunnel contrast are both wave phenomena by connecting the class wave definition to each part of the anchoring phenomenon. Their Running Wave Model (Lesson 2 entry) documents this first explanation attempt, and their 'still wondering' questions on the DQB identify the gaps — particularly around frequency, wavelength, and why different wave types behave differently in confined spaces like the Kabete tunnel — that subsequent lessons (3–12) will resolve.
Lesson 3	Using PhET: Wave on a String, students	The wave equation $v = f \lambda$ shows that	The wave equation gives us the numerical

The Wave Equation: Connecting Speed, Frequency, and Wavelength	observed that when frequency was increased while wave speed was held constant, the wavelength decreased in proportion, and that the product of frequency and wavelength remained approximately equal to the wave speed across all four data rows.	wave speed equals frequency multiplied by wavelength. For radio waves travelling at the speed of light, Classic 105 FM at 105.2 MHz has a wavelength of approximately 2.85 m, while a typical AM signal at 567 kHz has a wavelength of approximately 529 m — nearly 200 times longer.	tool to compare FM and AM waves. The very different wavelengths of FM (about 2.85 m) and AM (about 529 m) are our first quantitative clue to explain why the two signals behave differently in the Kabete tunnel. We cannot fully explain the tunnel mystery yet — that requires understanding diffraction in Lesson 6 — but we now have the numbers we need.
Lesson 4 Wave Types and Properties: Transverse, Longitudinal, and the Language of Waves	Students used the PhET Wave on a String simulation to observe transverse waves and watched a slinky demonstration of longitudinal waves. They measured wavelength using the on-screen ruler at multiple frequencies, recorded data in a structured table, tracked individual coil motion in the slinky using a tape marker, and posted evidence and new questions on the class Driving Question Board.	Students learned that waves are classified as transverse (particle oscillation perpendicular to wave travel, e.g., the FM radio signal from Classic 105) or longitudinal (particle oscillation parallel to wave travel, e.g., the sound from the matatu horn). They defined and applied amplitude, wavelength, frequency, period, and wave speed, and used $v = f\lambda$ to calculate that the Classic 105 FM signal has a wavelength of approximately 2.85 m while a 567 kHz AM signal has a wavelength of approximately 529 m.	The matatu horn produces a longitudinal mechanical wave — air particles compress and rarefy in the same direction as the sound travels from the matatu to the student's ear. The Classic 105 FM radio signal is a transverse electromagnetic wave with a wavelength of about 2.85 m. The dramatic difference in wavelength between FM (≈ 2.85 m) and AM (≈ 529 m) is now visible in the data, though the full explanation of why this causes the FM signal to disappear in the Kabete tunnel — involving wave diffraction around obstacles — remains an open question for the next lessons.
Lesson 5 The Wave Equation: Connecting Frequency, Wavelength, and Speed ($v = f\lambda$)	Using the PhET Sound Simulation, learners measured the wavelength of sound waves at three different frequencies (200 Hz, 400 Hz, 800 Hz) while speed remained constant (~ 340 m/s), and recorded the data in a three-column table. Using the LabXchange Wave Interference tool, learners observed that higher-frequency waves had shorter wavelengths. The class computed the ratios $\lambda_{200}/\lambda_{800}$ and f_{800}/f_{200} and found them equal, providing numerical evidence of the inverse relationship between frequency and wavelength at constant speed.	Learners derived the wave equation $v = f\lambda$ from first principles (number of waves per second $\times$ length of each wave = distance covered per second = speed). They applied the equation to calculate: (a) the wavelength of Classic 105 FM (105.2 MHz) ≈ 2.85 m; (b) the wavelength of KBC AM Radio (567 kHz) ≈ 529 m. They discovered that for a given wave speed, frequency and wavelength are inversely proportional — doubling the frequency halves the wavelength. They also connected these wavelengths to the size of the Kabete tunnel (~ 8 m) as a quantitative first step toward explaining the FM signal blackout.	The wave equation $v = f\lambda$ explains why Classic 105 FM and AM radio signals behave so differently inside the Kabete tunnel: FM waves at 105.2 MHz have a wavelength of only ~ 2.85 m, which is much smaller than the tunnel opening (~ 8 m), while AM waves at 567 kHz have a wavelength of ~ 529 m, far larger than the tunnel. This difference in wavelength relative to obstacle size is the quantitative foundation for the diffraction explanation to be completed in Lesson 6 — longer wavelengths diffract more readily around and through obstacles, while shorter wavelengths are more easily blocked.
Lesson 6 Diffraction of Waves: Why AM Survives the Kabete Tunnel	When waves pass through a gap in PhET Wave Interference, the waves spread out on the other side. The spreading (diffraction) is greatest when the gap is narrow, and is also greater when the wavelength is long. A very wide gap	Diffraction is the spreading of waves when they pass through a gap or around an obstacle. Diffraction is most pronounced when the gap width is similar to or smaller than the wavelength. Longer wavelengths (lower frequencies) diffract more through	The FM radio signal from Classic 105 (wavelength ~ 2.85 m) barely diffracts when it enters the Kabete tunnel opening (~ 8 m wide) because the gap is much larger than the FM wavelength — the signal travels nearly straight and most of its energy does

	produces almost no spreading — the waves continue almost straight through.	the same gap than shorter wavelengths (higher frequencies).	not bend around to reach the radio antenna inside the bus. The AM signal (wavelength ~529 m) diffracts very strongly because its wavelength is far larger than the tunnel opening — the AM signal bends widely around and into the tunnel, keeping the signal audible. This is why FM fades but AM survives in the Kabete tunnel.
Lesson 7 Interference of Waves: When Waves Meet, Add, and Cancel	Two wave sources in PhET Wave Interference produced a pattern of alternating strongly rippled regions and calm regions radiating outward from the sources. Strongly rippled points had path differences close to whole-number multiples of the wavelength. Calm points had path differences close to half-integer multiples of the wavelength. Increasing frequency made the pattern finer with more closely spaced nodal lines.	When two waves meet, the resultant displacement at any point is the algebraic sum of the individual displacements (principle of superposition). Constructive interference occurs where path difference = $n\lambda$ ($n = 0, 1, 2, \dots$), producing antinodal lines of maximum amplitude. Destructive interference occurs where path difference = $(n + 0.5)\lambda$, producing nodal lines of zero amplitude. These conditions are fixed in space for two coherent sources, creating a stable interference pattern.	The coloured rings on a soap bubble form because light reflects from both surfaces of the thin soap film. The two reflected waves have a path difference that depends on the local film thickness. Where the path difference matches a whole number of wavelengths of a particular colour, that colour undergoes constructive interference and is seen brightly. Where the path difference is a half-integer multiple of that wavelength, the colour is cancelled by destructive interference. Different colours have different wavelengths, so different colours are constructively reinforced at different thicknesses, producing the ring pattern. For the Kabete tunnel, FM waves that do diffract into the tunnel can also undergo destructive interference when reflected waves from the tunnel walls meet the incoming waves with a path difference of half a wavelength — adding a second mechanism to the FM signal loss first explained by diffraction in Lesson 6.
Lesson 8 Standing Waves: When a Wave Meets Itself	In the PhET simulation, when the frequency was set to specific values, the string stopped looking like a travelling wave and instead formed a pattern of sections — some parts moved up and down with large amplitude while other points never moved at all. When the frequency was increased to the next stable pattern, one more section appeared.	These stable patterns are called standing waves. They form when a wave reflects off the fixed end and superposes with the incoming wave. Points of permanent zero displacement are called nodes (permanent destructive interference) and points of maximum displacement are called antinodes (permanent constructive interference). For a string of length L , the wavelength of the n th harmonic is given by $\lambda_n = 2L / n$.	A guitar string vibrates in sections because only certain frequencies — those whose wavelength satisfies the condition that n half-wavelengths fit exactly into the string length — produce stable standing waves. At other frequencies, destructive interference prevents a stable pattern from building up. This same standing wave principle determines the length of FM and AM radio antennas: an FM dipole for Classic 105 at 105.2 MHz is about 1.4 m long, while an AM dipole for KBC at 567 kHz would be about 264 m long. This dramatic difference in scale is one more reason FM and AM waves behave so differently near tunnel

			boundaries — a thread we will complete in Lesson 12.
Lesson 9 Diffraction and Interference: Why Waves Bend Around Obstacles and Cancel or Reinforce Each Other	Using the LabXchange Wave Interference simulation, students observed that waves passing through a gap approximately equal to the wavelength spread out widely (diffract), while waves passing through a much larger gap continue with minimal spreading. Using the PhET Sound simulation with two in-phase speakers, students located audible dead-spots (regions of destructive interference) and loud regions (constructive interference) across the wavefront, recording at least three positions of each type in their evidence tables.	Diffraction is the bending of waves around obstacles or through gaps, and it is maximised when the gap width is approximately equal to the wavelength. The Principle of Superposition states that when two waves overlap, the resultant displacement at any point is the algebraic sum of the individual displacements — producing constructive interference (amplitude doubles, crests align) or destructive interference (amplitude cancels, crest meets trough). FM radio waves ($\lambda \approx 2.85$ m, $f = 105.2$ MHz) barely diffract through the Kabete tunnel mouth (~ 8 m wide) compared with AM waves ($\lambda \approx 529$ m), AND reflected FM waves inside the tunnel create destructive interference dead-zones — together explaining the FM signal blackout.	The FM radio signal on Classic 105 (105.2 MHz) cuts out in the Kabete tunnel for two compounding reasons: (1) FM wavelengths (~ 2.85 m) are smaller than the tunnel mouth width (~ 8 m), so the ratio $\text{gap}/\lambda \approx 2.8$, giving very limited diffraction — the signal barely bends into the tunnel; (2) inside the tunnel, the incoming FM signal and its reflections off the concrete walls overlap and produce destructive interference at many positions, creating dead-zones where the signal amplitude drops to near zero. AM waves ($\lambda \approx 529$ m) are unaffected because the tunnel mouth is tiny compared with their wavelength ($\text{gap}/\lambda \approx 0.015$), so AM waves diffract almost as if the tunnel wall were not there, and because their wavelength is vastly longer than the tunnel, reflected AM waves cannot form the fine pattern of interference dead-zones that affects FM, so a usable signal is maintained.
Lesson 10 Bending Light: Refraction, Snell's Law, and Total Internal Reflection	Students used the PhET Bending Light simulation to observe a laser ray bending at water–air and glass–air boundaries at five different angles of incidence. They also observed total internal reflection (TIR) when the angle of incidence exceeded the critical angle ($\sim 49^\circ$ for water, $\sim 42^\circ$ for glass). They connected the simulation to the canteen bent-straw supporting phenomenon.	Students discovered that the ratio $\sin\theta_i / \sin\theta_r$ is constant for a given pair of media — this constant is the refractive index n. They formalised Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$. They learned that TIR occurs when a wave travels from a denser to a less dense medium at an angle greater than the critical angle ($\sin\theta_c = 1/n$). They connected TIR to fibre-optic cables (TEAMS undersea cable, Mombasa) and linked refraction at boundaries to the Kabete tunnel FM signal loss.	Waves change speed when they cross a boundary between two media of different optical densities. This speed change causes the wave to bend (refract) — toward the normal when entering a denser medium, away from the normal when entering a less dense medium. When the angle is steep enough (beyond the critical angle), no energy crosses the boundary: total internal reflection occurs. The Kabete FM fade is explained by limited diffraction (FM $\lambda \approx 2.85$ m is small compared with the ~ 8 m tunnel opening), not by total internal reflection; TIR is the physics that keeps internet light signals travelling inside TEAMS fibre-optic cables all the way from Mombasa to the world.
Lesson 11 Resonance and Stationary Waves — Why the Tunnel Amplifies Some Sounds	Students observed standing wave patterns forming on a string in the PhET 'Wave on a String' simulation, identifying nodes (points of zero displacement) and antinodes (points of maximum displacement) at discrete	Stationary (standing) waves form when two waves of equal frequency and amplitude travel in opposite directions and superpose. In an enclosed space like a tunnel, the outgoing wave and its reflection interfere to	The selective behaviour of FM and AM signals in the Kabete tunnel is now explained by two wave behaviours working together: (1) INTERFERENCE / STATIONARY WAVES — FM wavelengths

and Kills Others	resonant frequencies. They also observed a live rubber-band demonstration in which only specific pluck frequencies produced sustained, symmetrical vibration patterns. The tunnel calculation revealed that FM wavelengths (~2.85 m for Classic 105) can satisfy $L = n\lambda/2$ for the ~120 m Kabete tunnel, while AM wavelengths (~529 m for KBC 567 kHz) cannot form a standing wave in the tunnel.	produce a standing wave only when the cavity length equals a whole number of half-wavelengths ($L = n\lambda/2$). This is called resonance. At resonance, energy builds up at antinodes; at other frequencies, destructive interference dominates. Nodes are points of permanent destructive interference; antinodes are points of permanent constructive interference. FM signals (short wavelength) can form interference patterns inside the Kabete tunnel, producing destructive dead-zones as well as reinforcement spots depending on position; AM signals (very long wavelength) cannot resonate inside the tunnel at all.	are short enough for reflections inside the tunnel to interfere, creating destructive dead-zones (nodes) where the signal is cancelled, as well as reinforcement spots (antinodes), depending on position. AM wavelengths are far too long to form any standing wave in the tunnel and so do not experience resonance-based cancellation or amplification; (2) DIFFRACTION (Lessons 6 and 9) — AM waves, having much longer wavelengths, diffract significantly around the tunnel entrance and exits, allowing them to 'bend in' and maintain some signal strength throughout; FM waves diffract less and are more directional. Together, these two mechanisms explain why Classic 105 FM fades inside the Kabete tunnel while the AM signal on the same receiver remains audible.
Lesson 12 Final Explanation — Pulling It All Together: From the Matatu Horn to the Kabete Tunnel	Students observed in the PhET Wave Interference simulation (single-slit mode) that an FM wave ($\lambda \approx 2.85$ m, simulating 105.2 MHz) passing through an 8 m gap (the Kabete tunnel) produces very little spreading beyond the gap, while an AM wave ($\lambda \approx 529$ m, simulating 567 kHz) passing through the same gap spreads almost uniformly in all directions. Students also observed their own Lesson 1 initial model diagrams — featuring vague or incorrect explanations — placed side by side with their detailed final annotated models, making conceptual change physically visible.	Students learned that ALL the wave behaviours encountered in this sub-strand — rectilinear propagation, reflection, refraction, diffraction, interference, resonance, and the Doppler effect — are not isolated facts but properties of a single underlying entity: a wave is a disturbance that transfers energy without transferring matter. They learned that the degree of diffraction depends on the ratio of wavelength to gap size (when $\lambda \approx$ gap or $\lambda \gg$ gap, significant diffraction occurs; when $\lambda \ll$ gap, minimal diffraction occurs), which is precisely why AM waves ($\lambda \sim$ hundreds of metres) bend around the Kabete tunnel walls while FM waves ($\lambda \sim 3$ m) do not. They learned that the Doppler effect — the apparent change in frequency due to relative motion between source and observer — explains why the matatu horn rises in pitch on approach and falls on departure, even though the driver never changes the horn. Together, these two explanations resolve both mysteries of the anchoring phenomenon.	Students explained in a collaboratively written scientific paragraph that: (1) The matatu horn appears to change pitch because of the Doppler effect — as the matatu approaches, successive sound wave crests are compressed into a shorter distance, increasing the frequency (and therefore pitch) detected at the student's ear; as it moves away, wave crests are stretched out, decreasing the detected frequency. The matatu's own horn frequency never changes — only the perceived frequency does. (2) The FM signal (Classic 105, 105.2 MHz, $\lambda \approx 2.85$ m) fades in the Kabete tunnel because FM waves have a wavelength far smaller than the tunnel opening (~8 m), meaning the ratio $\lambda/\text{gap} \ll 1$ and diffraction is minimal — FM waves travel in nearly straight lines and cannot bend around tunnel walls into the shadow zone. AM waves ($\lambda \sim$ hundreds of metres) have a wavelength far larger than the tunnel opening, so $\lambda/\text{gap} \gg 1$, strong diffraction occurs, and AM waves bend readily around and into the tunnel, reaching the receiver even in the shadow zone. Both explanations are unified by the same

			underlying principle: waves transfer energy according to properties — frequency, wavelength, amplitude — that govern how they interact with the physical world.
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